

Henry Liang

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Work Experience

Senior AI Engineer, Vultron AI – San Francisco, CA Apr 2025 – Present

- Shaped the core interaction model for an AI-native contract drafting platform, adapting human-agent interaction research into a steerable workflow for co-planning, co-execution, and ripple-aware replanning
- Launched a human-in-the-loop agent memory system that extracts negotiation ladders and learns client preferences across contract review rounds, decreasing lawyer review time by 45%
- Designed an agent eval harness to measure the contract review agent against RedlineBench, establishing reproducible baselines for regression testing, benchmark expansion, and standardized scoring
- Engineered a proposal ledger for AI contract review suggestions, enabling lawyer-controlled redline workflows, durable anchoring, version carry-forward, and clean artifact materialization
- Led the redesign of the LLM integration layer to introduce agentic capabilities, custom tooling, and model-provider-agnostic systems, reducing provider feature adoption time by 2 weeks
- Coordinated the on-call team to reduce CUDA OOM errors by 99% for customer deployments by redesigning OCR and reranker using model pools, custom configurations, and improved observability

Data Scientist II, Vail Systems, Inc. – Chicago, IL Feb 2024 – Apr 2025

- Led the development and fine-tuning of a Retrieval-Augmented Generation (RAG) system, achieving 99.18% top-5 retrieval accuracy to enhance enterprise question-answering capabilities.
- Spearheaded the engineering of an autonomous AI agent that analyzes daily call transcripts, extracts key insights, and generates executive summaries on call trends, sentiment drivers, and performance metrics
- Created a novel hybrid embedding that reduces the no-context rate for request-for-proposal completion by 83.33%, leading to a paper acceptance and invited talk at AI-ML Systems Conference, October 2024

Applied Scientist Intern, Amazon (Capstone) – Evanston, IL Sept 2023 – Dec 2023

- Designed an LLM application to perform graph reasoning for network routing anomaly detection, achieving 100% accuracy on core graph problems and scaling to graphs with 10K+ nodes
- Reduced erroneous graph database queries by 95% via prompt engineering and continuous fine-tuning of the query generation process using a few-shot learning feedback loop
- Co-authored paper on Graph Reasoning via Retrieval Augmented Framework (GRRAF), accepted for publication at EMNLP 2025

Machine Learning Researcher, Northwestern University – Evanston, IL Oct 2022 – June 2023

- Led design and implementation of a scalable AWS Internet of Things (IoT) streaming pipeline for real-time time-series forecasting; achieved a throughput of up to 55 predictions per second
- Trained and deployed an LSTM model on AWS EC2 for hourly Divvy bike demand forecasting for each station; realized a mean absolute error of 9 bikes

Skills

Languages: Python, TypeScript, JavaScript, SQL, Rust, R

Frameworks and Tools: Claude Code, Codex, Django, LangChain, Hugging Face, Git, Docker, Kubernetes

Education

Northwestern University – M.S. in Machine Learning and Data Science Dec 2023

University of California, Los Angeles – B.S. in Applied Mathematics, Statistics Mar 2022